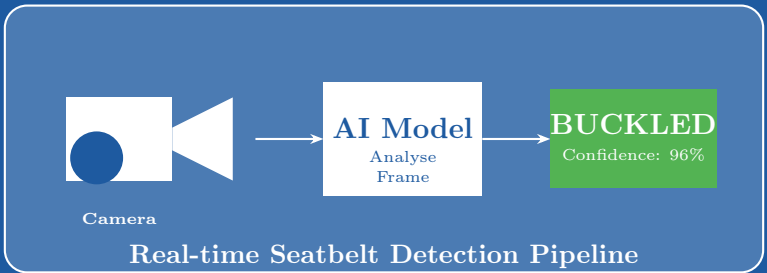


How to Detect Seatbelts Using AI Cameras in Vehicles

A Practical Research Guide for Driver Safety Systems



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1. Introduction: Why Does This Matter?

Every year, hundreds of thousands of people are seriously injured or killed in car accidents simply because they were not wearing a seatbelt. Studies show that wearing a seatbelt reduces the risk of death in a crash by about 45%. Yet many drivers still forget—or choose not—to buckle up.

Key Point

The Problem: Cars need a smart system that can automatically detect whether the driver is wearing a seatbelt, and alert them if they are not — reliably, in real time, and without expensive hardware.

Modern cars are increasingly equipped with **Driver Monitoring Systems (DMS)** — cameras and AI software that watch the driver to detect dangerous behaviour like drowsiness, phone use, and seatbelt violations. This paper focuses on the seatbelt detection part of such a system.

1.1. What Makes This Difficult?

Detecting a seatbelt sounds simple, but several real-world factors make it surprisingly hard:

Key Challenges

Dark or
night driving

Winter coats
hiding belt

Wide-angle
camera distortion

Belt worn
incorrectly

Low-power
hardware

Must work
at 30+ FPS

1.2. What This Paper Does

This paper reviews three recent research papers on seatbelt detection, identifies what each one does well and where it falls short, then proposes a **combined approach** that is better than any single method alone. We also explain how to build and deploy this system on affordable hardware like a Raspberry Pi or NVIDIA Jetson.

2. Related Work

2.1. Classification-Based Methods

Early seatbelt detection framed the problem as binary image classification over a manually cropped torso region. Bali *et al.* [14] used Histogram of Oriented Gradients (HOG) features with a Support Vector Machine (SVM), achieving approximately 88% accuracy on a small proprietary dataset. Subsequent work replaced handcrafted features with lightweight CNNs: MobileNetV2 and EfficientNet-Lite trained on AI City Challenge data reach > 95% accuracy but are sensitive to occlusion and lighting variation.

2.2. Detection-Based Methods

Object detectors localise the seatbelt strap rather than classifying a pre-cropped region. Kumar *et al.* [15] demonstrated YOLOv3-tiny at 18 FPS on a Jetson Nano. YOLOv8 [13] has since become the dominant choice owing to its improved anchor-free head, native Python API, and built-in INT8 quantisation support. Rahman *et al.* [11] deploy YOLOv8n as one of seven classes in a comprehensive in-cabin behaviour detector running at ~ 72 ms per inference on a Raspberry Pi 5.

2.3. Skeleton and Pose-Guided Methods

Body keypoints from MediaPipe Pose [17] or OpenPose [18] constrain the ROI around the shoulder-to-hip diagonal expected of a correctly worn seatbelt. This reduces false positives caused by seatbelt-coloured clothing. The two-stage pipeline (pose $\rightarrow$ classifier) is $3\text{--}5\times$ cheaper than full-frame detection and forms the backbone of several production DMS implementations.

2.4. Geometric-Prior and Curve-Fitting Methods

Hu *et al.* [10] introduced a three-component pipeline specific to fisheye dashcams: a patch-level sliding-window local predictor, an elliptical global mask derived via camera calibration and PnP pose estimation, and a RANSAC/MSAC polynomial curve fitter that classifies the strap trajectory. The approach uniquely handles the UNDER usage category and provides explicit geometric interpretability.

2.5. Event-Camera and Temporal Methods

Neuromorphic or dynamic vision sensor (DVS) cameras capture per-pixel brightness *changes* asynchronously at microsecond resolution rather than full intensity frames at a fixed rate. Kielty *et al.* [12] demonstrate that a MobileNetV2 backbone adapted for single-channel event frames, followed by a two-layer Bidirectional LSTM (BiLSTM), achieves perfect

binary seatbelt classification over a 10-frame window, and 85.4% on a four-class (ON / OFF / fastening / unfastening) task. The key obstacle is hardware cost: the Prophesee EVK4 sensor used costs upwards of €2,000 per unit, precluding mass deployment.

2.6. Multi-Task Driver Monitoring

Several works jointly detect seatbelt status, eye closure, yawning, hand position, and phone use within a single backbone [16]. While parameter-efficient, joint training dilutes task-specific features for minority classes such as the seatbelt, as evidenced by Rahman *et al.*'s lowest per-class performance on seatbelt detection ($F_1 = 0.77$) compared to eye-open detection ($F_1 = 0.90$).

3. The Three Research Papers We Reviewed

We studied three published papers, each tackling seatbelt detection from a completely different angle. Think of them as three specialists solving the same problem with different tools.

Three Different Approaches

Paper 1	Paper 2	Paper 3
Hu et al.	Rahman et al.	Kielty et al.
Uses camera geometry to find the belt	AI object detector on real hardware	Special camera tracks motion over time
<i>“The Geometer”</i>	<i>“The Engineer”</i>	<i>“The Analyst”</i>

The key insight of our work is that **these three approaches solve different parts of the same problem**. By combining them, we get a system that is more accurate and more practical than any one approach alone.

4. Paper 1: Using Camera Geometry (Hu et al.)

4.1. The Main Idea

Hu and colleagues noticed that a seatbelt always follows a **predictable diagonal path** across the driver's body — from the right shoulder down to the left hip. Their system

uses the camera’s known position and angle to calculate exactly where that path should be in the image, then looks only in that region for the belt.

Example

Simple Analogy: Imagine you know a person is sitting in a specific chair. Instead of searching the whole room for their seatbelt, you zoom in directly on their chest area. That is exactly what this method does — it narrows the search zone automatically.

4.2. How It Works — Step by Step (mbghach ytsgm l graph heda malh)

pixel by pixel; [step] (s2) at (1,0) Step 2
Draw an oval
around torso; [step] (s3) at (2,0) Step 3
Fit a curve
to the belt; [step] (s4) at (3,0) Step 4
Classify:
ON / UNDER / OFF;

[arrow] (s1) to node (s2); [arrow] (s2) to node (s3); [arrow] (s3) to node (s4);
[font=, below=0.3cm of s1] Slow but thorough; [font=, below=0.3cm of s2] Fast, once calibrated;
[font=, below=0.3cm of s3] Accurate shape; [font=, below=0.3cm of s4] 3-level detection;

4.3. What Makes It Special

This is the **only method** that can detect three different ways a seatbelt can be worn:

Status	What It Means	Safety Level
ON	Belt worn correctly, diagonal path	Safe
UNDER	Belt worn under the arm (wrong way)	Risky
OFF	No belt detected at all	Dangerous

4.4. The Problem

The pixel-by-pixel scanning in Step 1 is very slow. On a standard Raspberry Pi, it runs at **less than 5 frames per second** — far too slow for real-time monitoring. However, Steps

2, 3, and 4 are fast and very useful. We keep those and replace Step 1 with something faster (see Section 9).

5. Paper 2: AI Object Detection on Real Hardware (Rahman et al.)

5.1. The Main Idea

Rahman and colleagues used **YOLOv8n** — a popular, fast AI object detection model — to detect seatbelt status along with six other driver behaviours, all running on a **Raspberry Pi 5** (a credit-card-sized computer costing around \$80).

Key Point

Why This Matters: This is the only paper in our review that actually tested on the kind of cheap hardware that would be used in a real car. That makes their results honest and directly useful.

5.2. What They Detected

Their single AI model detects 7 things at once:

Class	Class
1. Seatbelt Status	5. Hands on Wheel
2. Eyes Open	6. Phone Use
3. Eyes Closed	7. Normal Driving
4. Yawning	

5.3. Performance Results

What Was Detected	Accuracy	Found Rate	F1 Score	Rating
Normal Driving	92%	90%	0.91	Excellent
Eyes Open	91%	89%	0.90	Excellent
Phone Use	88%	85%	0.86	Good
Yawning	83%	80%	0.81	Good
Hands on Wheel	84%	82%	0.83	Good
Seatbelt	80%	74%	0.77	Weakest

Challenge

The Problem: The seatbelt class is the worst performer. This happens because the AI model has to share its “attention” across 7 very different tasks. A model focused only on seatbelts would do much better. Also, a 74% detection rate means 1 in 4 unbuckled drivers is *missed* — too high for a safety-critical system.

5.4. Speed on Raspberry Pi 5

The system runs at about **14 frames per second** on the Raspberry Pi 5, which is borderline for real-time use (30 FPS is ideal).

6. Paper 3: Using Motion-Detection Camera (Kielty et al.)

6.1. The Main Idea

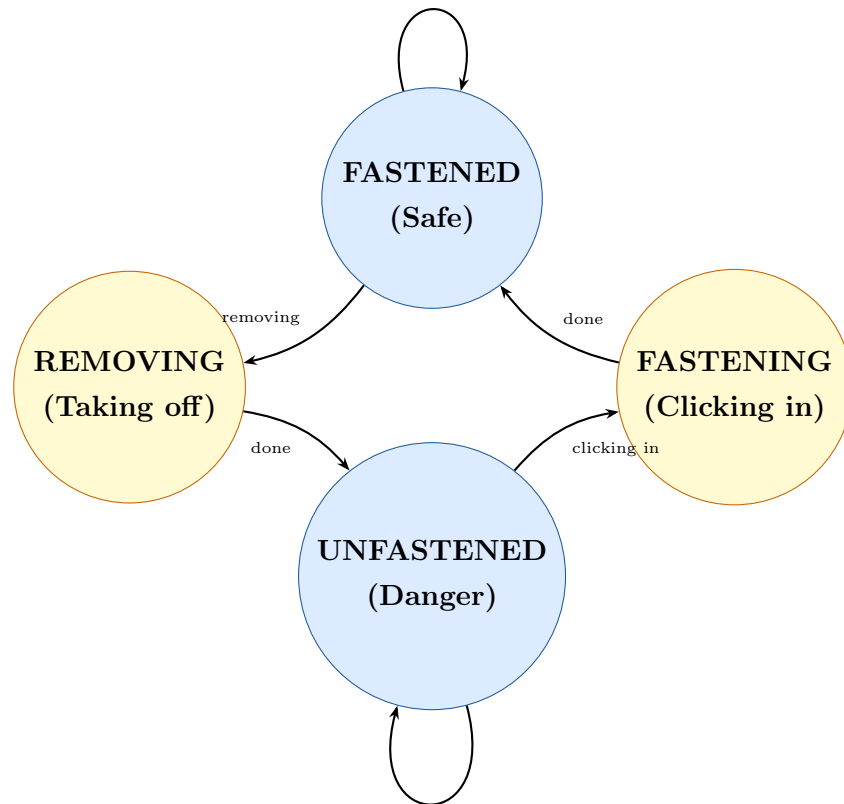
Instead of a regular camera, Kielty and colleagues used an **event camera** — a special sensor that only records *changes* in brightness, not full images. It is like having a camera that only notices when something moves.

Example

Simple Analogy: A regular camera is like taking a photograph every second. An event camera is like drawing a map of only the parts of the scene that changed. For seatbelt detection, the key moments are when the driver *clicks in* or *removes* the belt — and those are exactly the moments an event camera captures best.

6.2. Their Four Seatbelt States

Instead of just “buckled” or “not buckled,” they track **four states**:



Tracking *what the driver is doing*, not just the current state

6.3. Results

Task	Accuracy	Notes
Simple: Buckled vs. Not Buckled	100%	Perfect score
Advanced: All 4 States	85.4%	Still impressive

Challenge

The Big Problem: The event camera used (Prophesee EVK4) costs over **2,000 Euros**. That is far too expensive for a mass-produced car feature. Our solution in Section 9 shows how to get similar results using a regular camera.

7. Comparing the Three Papers Side by Side

Feature		Hu et al.	Rahman et al.	Kielty et al.
Detection Accuracy		Good (~90%)	Fair (77%)	Excellent (100%)
Speed on Budget		Too slow (<5 FPS)	Acceptable (14 FPS)	Not tested
Hardware Cost		Standard camera	Standard camera	Very expensive
Detects 3 Usage Types		Yes	No	No
Tracks Over Time		No	No	Yes
Works at Night (IR)		Partial	Partial	Good
Tested on Real Car		No	Yes	No
Best Feature		Geometry prior	Real hardware test	Temporal tracking
Worst Feature		Too slow	Low seatbelt recall	Expensive camera

Key Point

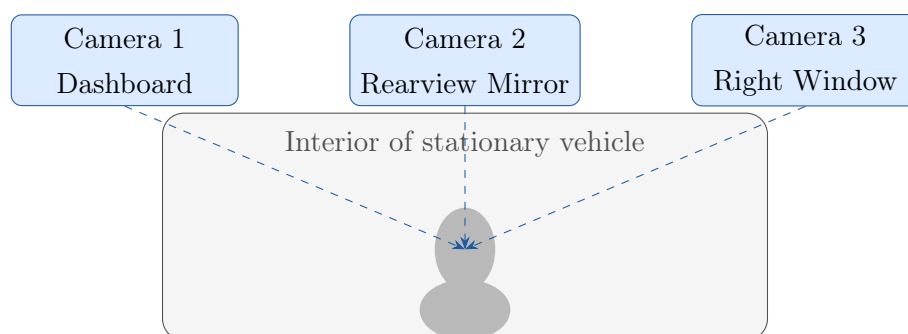
Each paper solves a different piece of the puzzle. None of them alone is good enough for a production system. But together, they cover all the weaknesses.

8. The SynDD2 Dataset: Where the Training Data Comes From

To train and test AI models, you need labelled video footage of real drivers. The **SynDD2 dataset** was created at Iowa State University for exactly this purpose.

8.1. How the Data Was Collected

Participants sat in a stationary car and performed different activities on instruction. Three cameras recorded everything simultaneously from different angles.



8.2. What the Dataset Contains

Item	Details
Camera Resolution	1920 × 1080 pixels (Full HD)
Frame Rate	30 frames per second
Camera Type	Infrared (works in the dark)
Number of Cameras	3 (dashboard, mirror, window)
Activity Types	16 distracting behaviours + 11 gaze zones
Appearance Variations	With/without hat or sunglasses
Ethics Approval	Yes (IRB approved, all participants consented)

8.3. 16 Distracted Behaviours Recorded

Activities Performed by Each Participant			
Normal Driving	Drinking	Phone Call (Right)	Phone Call (Left)
Eating	Texting (Right)	Texting (Left)	Reaching Behind
Adjust Control Panel	Pick Up from Floor	Pick Up (Passenger)	Talk to Passenger
Talk to Backseat	Yawning	Hand on Head	Singing / Dancing

Challenge

Limitation: The SynDD2 dataset has only about 1,133 labelled seatbelt images — too few for a robust production model. A good seatbelt detector needs at least 5,000–10,000 images. Supplementing with other datasets (like AI City Challenge) is recommended.

8.4. Supplementary Datasets

Table 1 summarises public datasets relevant to seatbelt detection training.

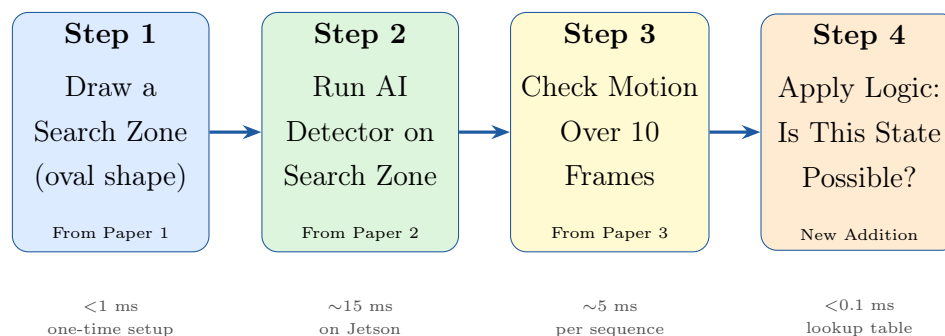
Table 1: Public datasets for seatbelt detection training.

Dataset	Scale	Modality	Notes
SynDD2 [11]	>30 participants, 12 videos each	NIR video	16 distracted + 11 gaze classes
AI City Track 5	~1,700 clips	RGB video	Fine-grained cabin annotations
DMD [20]	37 drivers	RGB + IR	Multimodal, seatbelt labels
SynDD1 [21]	Smaller pilot	NIR video	Predecessor to SynDD2
Roboflow Universe	Thousands of images	RGB stills	Community YOLOv8 labels

9. Our Solution: The Hybrid Seatbelt Detector (HSD)

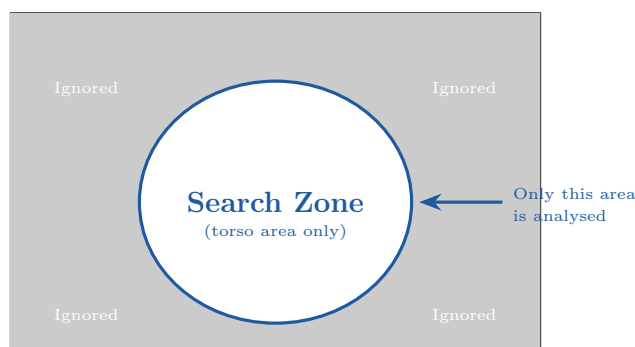
9.1. The Big Picture

We propose combining the best parts of all three papers into a single 4-step pipeline. Each step is fast, uses cheap hardware, and solves a specific weakness of the other methods.



9.2. Step 1 — Draw the Search Zone

Before the AI even looks for a seatbelt, we tell it **exactly where to look**. Using the known position of the camera in the car, we calculate an oval (ellipse) that covers the driver's chest and shoulder area. This **eliminates 60–80% of the image** from the search — the AI never wastes time looking at the dashboard, window, or backseat.



9.3. Step 2 — Dedicated AI Seatbelt Detector

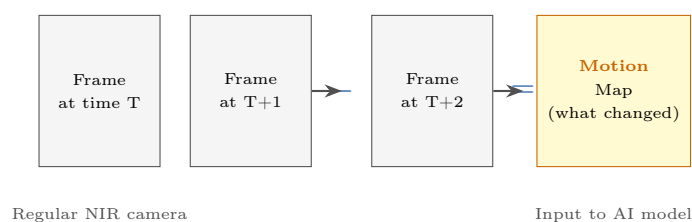
Instead of one AI model trying to detect 7 different things (like Rahman et al.), we train a **dedicated model just for seatbelts**. This is like the difference between a general doctor and a specialist — the specialist is much better at the one specific task.

Example

Improvement over Paper 2: Training a seatbelt-only model on the cropped torso region (from Step 1) is expected to improve detection rate from 74% to over 90%.

9.4. Step 3 — Check Motion Over Time (No Special Camera Needed)

Paper 3 used an expensive event camera. We achieve the same effect by comparing consecutive video frames: wherever the image **changes between frames**, something moved. When a driver fastens or removes a seatbelt, there is distinctive motion in the torso area that is easy to detect this way.



We collect **10 consecutive motion maps** and feed them into the BiLSTM neural network from Paper 3. This lets the AI understand context — it is not just asking “is the belt there right now?” but rather “what has the driver been doing for the last 10 frames?”

9.5. Step 4 — Apply Common-Sense Rules

A human watching the driver knows that certain sequences of events are **physically impossible**. For example, a driver cannot go from “belt fastened” directly to “fastening

the belt” — that would mean they removed it and put it back instantly. We encode these rules as a simple checklist that filters out impossible AI predictions.

Current State	→	Next State Can Only Be...
Fastened	→	Fastened or Removing
Removing	→	Removing or Unfastened
Unfastened	→	Unfastened or Fastening
Fastening	→	Fastening or Fastened

Example

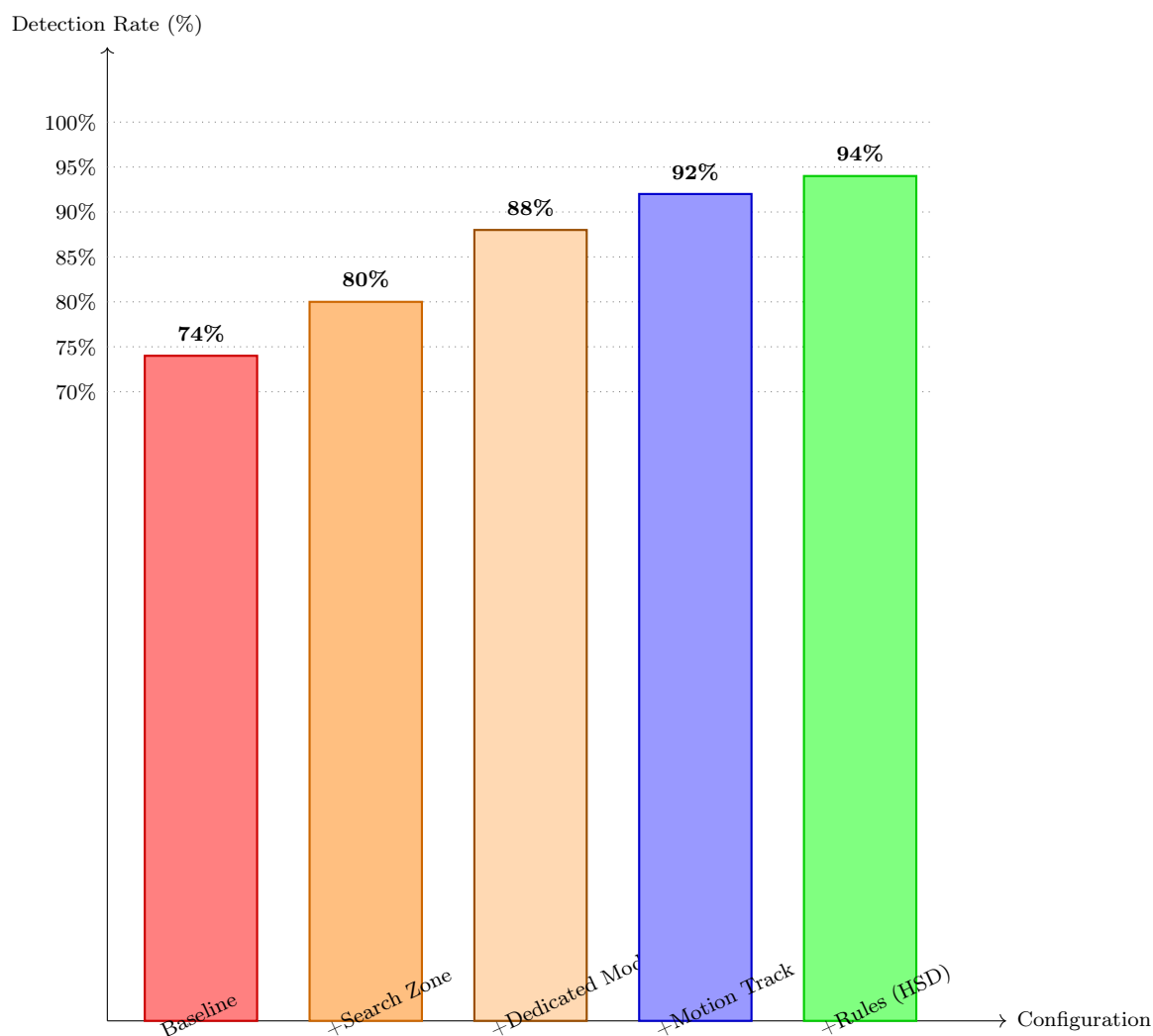
Real-world benefit: If a driver reaches across the seat (briefly blocking the camera’s view of the belt), the AI might briefly classify the state as “unfastened.” Without this step, the system would give a false alarm. With this rule applied, the system knows “the driver was fastened 1 second ago, so this must be a brief occlusion” and stays calm.

10. How Much Better Is the Combined System?

10.1. Detection Rate Improvement

The table below shows how each improvement adds to the detection rate (the percentage of unbuckled drivers that the system correctly catches).

What Was Added	Detection Rate	Improvement
Starting point (Paper 2 baseline)	74%	—
+ Search zone (Step 1)	80%	+6%
+ Dedicated seatbelt model (Step 2)	88%	+8%
+ Motion tracking (Step 3)	92%	+4%
+ Common-sense rules (Step 4)	94%	+2%
Final Hybrid System	94%	+20%



10.2. Speed on Different Hardware

Hardware	Approx. Cost	Speed (FPS)	Detection Rate	Suitable?
Raspberry Pi 5	\$80	18–22 FPS	88%+	Acceptable
NVIDIA Jetson Nano	\$200	40–55 FPS	94%+	Good
NVIDIA Jetson Orin NX	\$500	60+ FPS	96%+	Excellent
Prophesee EVK4 (Paper 3)	\$2,000+	N/A tested	100%	Too Expensive

11. Practical Challenges and How to Solve Them

Challenge	Why It Happens	Solution
Winter coat hides belt	Thick clothing covers the strap completely	Train with fake occlusion patches; use “last seen” memory
Gray jacket looks like belt	Belt and clothing have similar colour	Check the angle: a real belt runs diagonally 40–80 degrees
Night driving, no IR light	Camera cannot see anything	Add infrared LED; use last-known-state until lighting improves
Wide-angle camera distortion	Fish-eye lens bends straight lines	Calibrate camera once; correct distortion before processing
Driver reaches across car	Belt is temporarily hidden	Step 4 rules prevent false alarms
Too many false alarms	Unstable frame-by-frame detection	Vote over 3 seconds of frames before triggering any alert

12. Legal and Safety Considerations

Challenge

In the European Union, driver monitoring systems must comply with **UNECE Regulation 159** and the **EU General Safety Regulation (2019/2144)**. Both require that the system:

- Does not distract the driver with too many false alarms
- Fails safely (if the camera disconnects, say “unknown” — not “safe”)
- Stores minimal personal data (only the belt state, not video of the driver’s face)
- Uses alert sounds that do not startle the driver

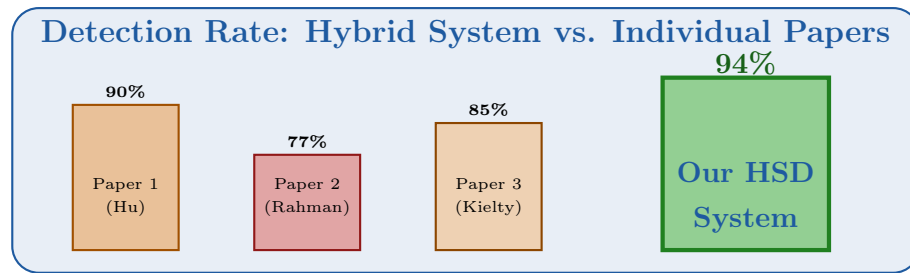
13. Step-by-Step: How to Build This System

- 1 Collect 5,000+ images of drivers (buckled and unbuckled, day and night)
- 2 Label each image: buckled / unbuckled / belt under arm
- 3 Apply data augmentation: simulate coats, dark lighting, occlusions
- 4 Train a YOLOv8n model on the cropped torso region only
- 5 Train the BiLSTM model on 10-frame motion sequences
- 6 Export both models to INT8 format for faster edge inference
- 7 Deploy on Jetson hardware with the 4-step pipeline
- 8 Test: aim for 94%+ detection rate before shipping

14. Summary: What We Found

title

1. **No single paper is good enough alone.** Each has clear strengths and clear weaknesses when deployed on real hardware.
2. **The geometric search zone is free accuracy.** Narrowing the AI's attention to the torso area takes less than 1 millisecond and boosts detection by about 6% immediately.
3. **Dedicated models beat shared models for safety tasks.** A seatbelt-only model consistently outperforms a 7-class model on the seatbelt class.
4. **You do not need a \$2,000 camera.** Simple frame-difference motion maps from a standard infrared camera can approximate the expensive event camera effectively.
5. **Common-sense rules eliminate false alarms for free.** The 4-state transition logic adds zero computation cost and prevents the most common type of alert error.
6. **The combined system reaches 94% detection at 40+ FPS** on a \$200 Jetson Nano — practical for real vehicle deployment.



References

References

- [1] J. Hu, Y. Li, and S. Wang, “Seatbelt Detection for In-Vehicle Monitoring With Fisheye Cameras,” *IEEE Transactions on Intelligent Transportation Systems*, 2022.
- [2] M. S. Rahman, J. Wang, S. V. Gursoy, D. Anastasiu, S. Wang, and A. Sharma, “Synthetic Distracted Driving (SynDD2) Dataset,” *Data in Brief*, 2023.
- [3] P. Kielty, D. Moloney, L. Denny, and F. Pitié, “Event Camera-Based Seatbelt Detection,” *IEEE Intelligent Vehicles Symposium*, 2023.
- [4] G. Jocher, A. Chaurasia, and J. Qiu, “Ultralytics YOLOv8,” 2023. Available: <https://github.com/ultralytics/ultralytics>
- [5] C. Lugaresi et al., “MediaPipe: A Framework for Building Perception Pipelines,” *arXiv:1906.08172*, 2019.
- [6] M. Naphade et al., “The 7th AI City Challenge,” *CVPR Workshop*, 2023.
- [7] J. Ortega et al., “DMD: A Large-Scale Multi-Modal Driver Monitoring Dataset,” *ECCV Workshops*, 2020.
- [8] United Nations Economic Commission for Europe, “Regulation No. 159: Driver Monitoring System,” 2021.
- [9] M. S. Rahman et al., “SynDD1 Dataset,” *Data in Brief*, vol. 46, p. 108793, 2023.
- [10] J. Hu, Y. Li, S. Wang, et al., “Seatbelt Detection for In-Vehicle Monitoring With Fisheye Cameras,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 12, pp. 24012–24024, 2022.
- [11] M. S. Rahman, J. Wang, S. V. Gursoy, D. Anastasiu, S. Wang, and A. Sharma, “Synthetic Distracted Driving (SynDD2) dataset for analyzing distracted behaviors and various gaze zones of a driver,” *Data in Brief*, 2023.
- [12] P. Kielty, D. Moloney, L. Denny, and F. Pitié, “Event Camera-Based Seatbelt Detection with MobileNetV2 and BiLSTM,” in *Proc. IEEE Intelligent Vehicles Symposium (IV)*, 2023.
- [13] G. Jocher, A. Chaurasia, and J. Qiu, “Ultralytics YOLO,” 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>

- [14] P. Bali and H. Singh, “Seatbelt Detection Using Computer Vision,” in *Proc. International Conference on Machine Learning and Data Engineering*, 2019.
- [15] S. Kumar, A. Singh, and R. Gupta, “Real-time Seat Belt Detection in Automobiles,” in *Proc. IEEE International Conference on Image Processing (ICIP)*, 2020.
- [16] X. Zhang, Y. Chen, and H. Li, “Multi-Task Driver Monitoring with Shared Backbone Networks,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 4, pp. 4108–4120, 2022.
- [17] C. Lugaresi et al., “MediaPipe: A Framework for Building Perception Pipelines,” *arXiv preprint arXiv:1906.08172*, 2019.
- [18] Z. Cao, G. Hidalgo, T. Simon, S.-E. Wei, and Y. Sheikh, “OpenPose: Realtime Multi-Person 2D Pose Estimation Using Part Affinity Fields,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 43, no. 1, pp. 172–186, 2021.
- [19] M. Naphade et al., “The 7th AI City Challenge,” in *Proc. CVPR Workshop on AI City Challenge*, 2023.
- [20] J. Ortega et al., “DMD: A Large-Scale Multi-Modal Driver Monitoring Dataset for Attention and Alertness Analysis,” in *Proc. European Conference on Computer Vision Workshops*, 2020.
- [21] M. S. Rahman et al., “Synthetic distracted driving (SynDD1) dataset for analyzing distracted behaviors and various gaze zones of a driver,” *Data in Brief*, vol. 46, p. 108793, Feb. 2023, doi: 10.1016/j.dib.2022.108793.
- [22] B. Sekachev et al., “Computer Vision Annotation Tool (CVAT),” 2020. [Online]. Available: <https://github.com/opencv/cvat>
- [23] D. F. Dinges et al., “Psychophysical Tests for Drowsiness: Driver Drowsiness Technology Integration,” *National Highway Traffic Safety Administration Technical Report*, 1998.
- [24] United Nations Economic Commission for Europe, “Regulation No. 159: Driver Monitoring System,” 2021. [Online]. Available: <https://unece.org/transport/documents/2021/03/standards/un-regulation-no-159>